

Surname : SOLUTION GUIDE

Initials :

Student number:

STICK BARCODE STICKER HERE

FSK 116

Semester Test 2

Department of Physics, University of Pretoria

Date: 2025-05-06

Time: 3.0 hours

Total: 75

Examiner: RQ Odendaal

Moderator: Prof JM Nel

%< ✂ ✂ ✂ ✂

Test register: Tear carefully on the line above and hand to the invigilator.

All exam regulations are applicable to this test as indicated in point 8 on the next page.

FSK 116: Semester Test 2

2025-05-06

Surname					Initials		
Student number							
Venue							
Signature							

STICK BARCODE STICKER HERE

1. Answer all the questions in the space provided on the question paper.
2. No pencil work will be marked. Do not use "tippex".
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues.
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams).
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations!**
A correct answer does not imply that you will receive full marks.
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. The use of the cheat sheet is only allowed subject to the rules stated on it.
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

Constants and formulas

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$;
 $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J/(mol}\cdot\text{K)}$; $N_A = 6.02 \times 10^{23} / \text{mol}$;
 $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 299\,792.5 \text{ km/s}$; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$;
 $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$; $\rho_{\text{water}} = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = 6.37 \times 10^6 \text{ m}$; $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

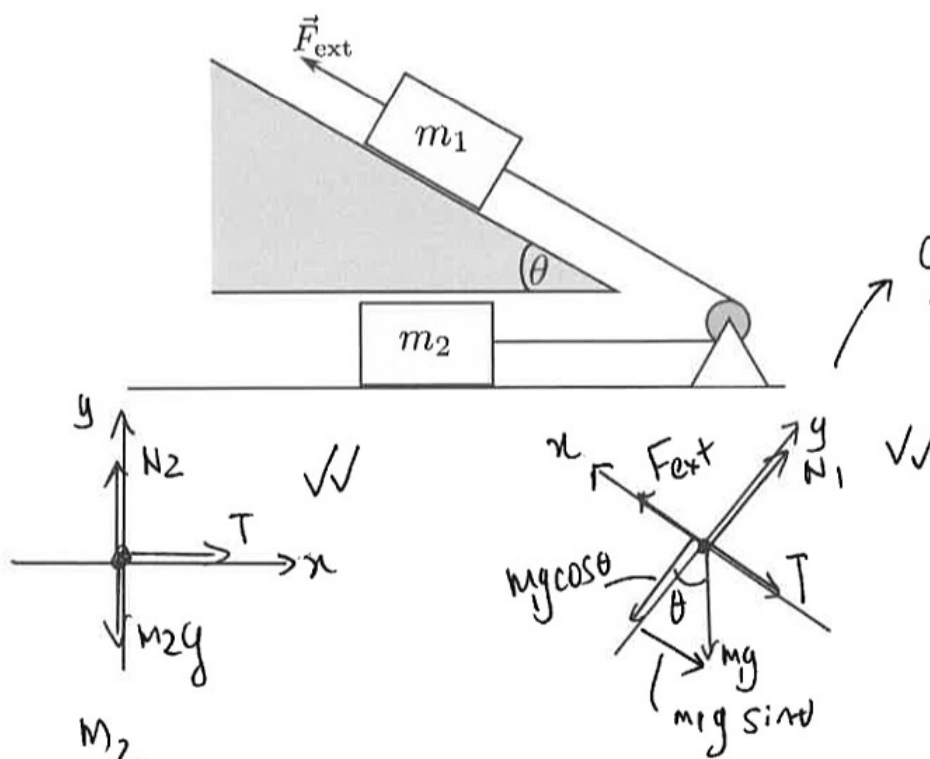
1 Question 1

[10]

The figure shows an object of mass m_1 on an inclined surface with an angle of θ above the horizontal. The object m_1 is connected to a second object of mass m_2 on a horizontal surface below an overhang that is formed by the inclined surface. An external force, $\vec{F}_{\text{ext}}$, is exerted on the object with mass m_1 . Both objects are observed to accelerate. Assuming the surfaces and the pulley are frictionless, and the pulley and the connecting string are massless, show that the tension in the string is

$$T = \frac{m_2}{m_1 + m_2} (F_{\text{ext}} - m_1 g \sin \theta).$$

You must use the component method together with the appropriate Newton law and free-body diagrams to solve the problem.



4

b

We assume a common acceleration in the $+x$ -direction for both blocks. $\Rightarrow a_{1x} = a_{2x} = a$. The tension is also common for both blocks, label it T .

For m_1 : $\sum F_x = m_1 a$ ✓

$$F_{\text{ext}} - T - m_1 g \sin \theta = m_1 a \quad \text{--- (1)}$$

For m_2 : $T = m_2 a$ ✓

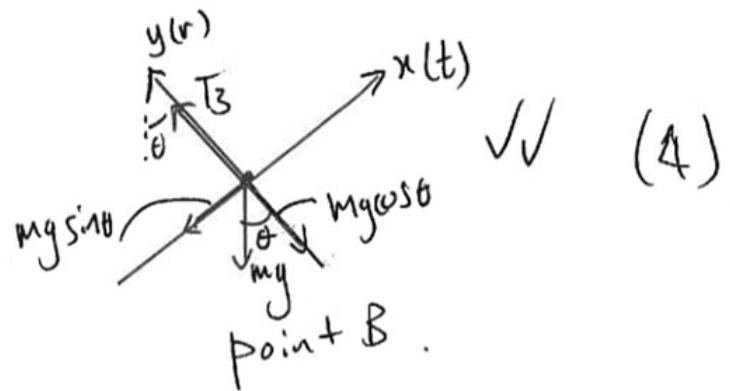
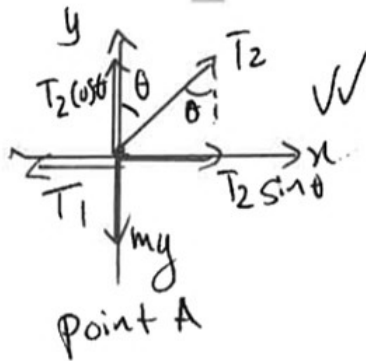
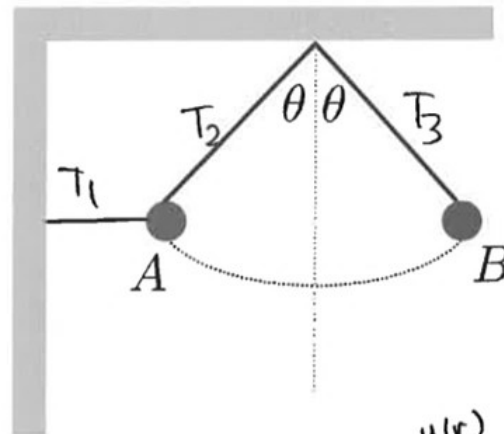
$$a = \frac{T}{m_2} \quad \text{--- (2)} \quad \checkmark$$

Eliminate a ✓ to find the expression asked for. ✓

2 Question 2

[8]

A ball is held at rest at position A in the figure by two light cords. The horizontal cord is cut, and the ball starts swinging as a pendulum. Point B is the farthest to the right that the ball goes as it swings back and forth. What is the ratio of the tension in the supporting cord at position B to what it was at A before the horizontal cord was cut?



At point A : y-component : $\sum F_y = 0$

$$T_2 \cos \theta - mg = 0$$

$$T_2 = \dots$$

At point B : Radial component : $\sum F_r = \frac{mv^2}{R}$ but $v=0$

$$T_3 - mg \cos \theta = 0$$

$$T_3 = \dots$$

turning point

$$\frac{T_3}{T_2} = \dots = \cos^2 \theta$$

(4)

3 Question 3

[7]

The moon's radius is 1.74×10^6 m, and the gravitational acceleration of the moon is 1.63 m/s^2 . A spaceship is orbiting just above the moon's surface.

- (a) Find the speed of the spaceship as it orbits the moon.

(5)

5

$$a_c = \frac{v^2}{R_m} = \frac{GM_m}{R_m^2} = g_m$$

$$\Rightarrow v = \sqrt{g_m R_m} = \dots = 1.68 \text{ km/s}$$

- (b) What is the period of the spaceship as it orbits the moon?

(2)

2.

$$\text{Circumference of the orbit: } d_2 = 2\pi R_m$$

$$= \dots$$

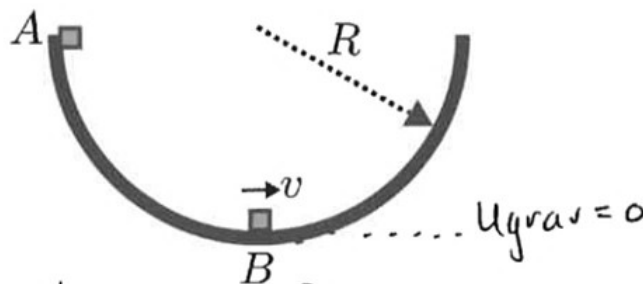
$$= 1.09 \times 10^4 \text{ km}$$

$$T = \frac{d}{v} = \dots = 108 \text{ min}$$

4 Question 4

[6]

An object of mass 0.1 kg is released from rest at point A , which is at the top edge of a hemispherical bowl of radius $R = 0.4 \text{ m}$. When it reaches point B at the bottom of the bowl, the object has a speed of $v = 1.8 \text{ m/s}$. Calculate the work done by friction on the object when it moves from A to B . Clearly explain all choices made for initial and final values.



$$\Delta K + \Delta U_{\text{grav}} + \Delta E_{\text{int}} = 0$$

$$K_f - K_i + U_{\text{grav},f} - U_{\text{grav},i} + \Delta E_{\text{int}} = 0$$

6

$$\frac{1}{2}mv^2 - 0 + 0 - mgyR = -\Delta E_{int}$$

$\uparrow$ Start from rest
 $\nwarrow$ Chose it to be zero at point B.

$$0.162 \text{ J} - 0.392 \text{ J} = -\Delta E_{int}$$

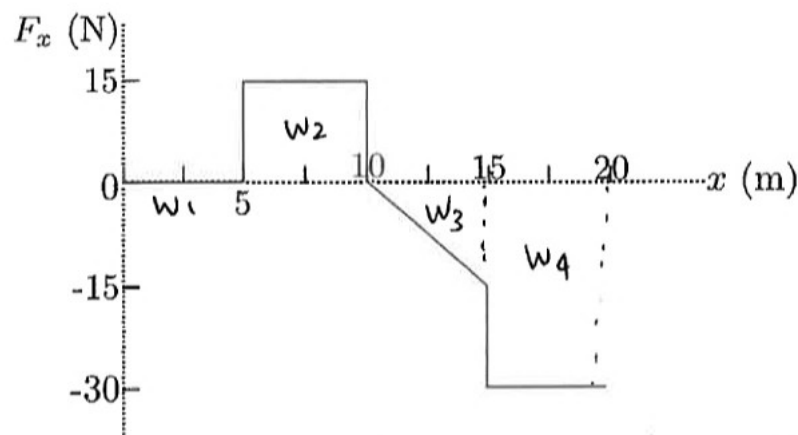
$$\Rightarrow W_{fric} = \Delta E_{int} = 0.230 \text{ J}$$

5 Question 5

[5]

5

A varying force is exerted on a particle as it moves in the positive direction along the x -axis. The figure shows a graph of this force F_x versus the particle's position x . Find the work done by this force on the particle as the particle moves from $x_i = 0$ to $x_f = 20$ m.



Calculate area under the curve for each part indicated (show calculation!) ✓

$$W_1 = 0 \quad W_2 = +75 \text{ J} \quad W_3 = -37.5 \text{ J} \quad W_4 = -150 \text{ J}$$

$$W_{net} = W_1 + W_2 + W_3 + W_4 = \dots = -112.5 \text{ J} \checkmark$$

6 Question 6

[6]

A single object is displaced by a net force $\Sigma \vec{F}$ acting only in the x -direction. We know from Newton's second law that the object moves with an acceleration $\vec{a}$. If the object moves through a displacement $\Delta \vec{r} = \Delta x \hat{i}$, the net work done on the object by the net force is

$$W_{\text{net}} = \int_{x_i}^{x_f} \Sigma F dx.$$

Use this result to prove the work-kinetic energy theorem that is $W_{\text{net}} = \Delta K$, where K is the kinetic energy of the object.

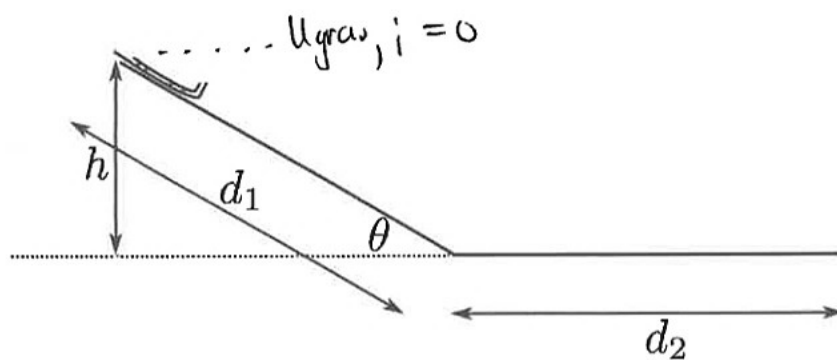
Check class notes or page 161 of Serway
(equations 7.14 to 7.17 of Serway)

7 Question 7

[9]

A child on a sled starts from rest and slides down a snow-covered hill. The hill's slope makes an angle $\theta = 35^\circ$ with the horizontal. The surface of the hill is $d_1 = 25.0$ m long. When the boy and the sled reach the bottom of the hill, they continue sliding on a horizontal snow-covered field. The coefficient of kinetic friction between the sled and the snow is $\mu_k = 0.100$.

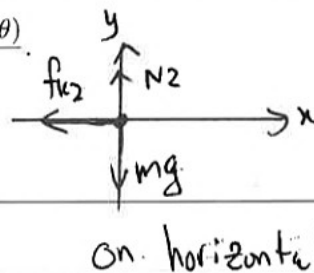
Clearly explain all choices made for initial and final values, even when they are zero.



(a) Show that the horizontal distance d_2 the sled and child travels at the bottom of the hill is given by



$$d_2 = \frac{d_1(\sin \theta - \mu_k \cos \theta)}{\mu_k}$$



(8)

8

$$f_{k1} = \dots = \mu_k m g \cos \theta$$

$$f_{k2} = \dots = \mu_k m g$$

Work done by friction : $W_{fric} = -f_{k1}d_1 - f_{k2}d_2$

$$= -\mu_k m g \cos \theta d_1 - \mu_k m g d_2$$

Start and end at rest

$$\Delta K + \Delta U_g + \Delta E_{int} = 0$$

$$0 + U_{grav,f} - U_{grav,i} = -\Delta E_{int}$$

$$0 - mgh - 0 = -f_{k1}d_1 - f_{k2}d_2 = W_{fric}$$

$$-m g d_1 \sin \theta$$

choice of where U_{grav} is zero

$$\text{find } d_2 = \dots$$

(b) What is the value of d_2 ?

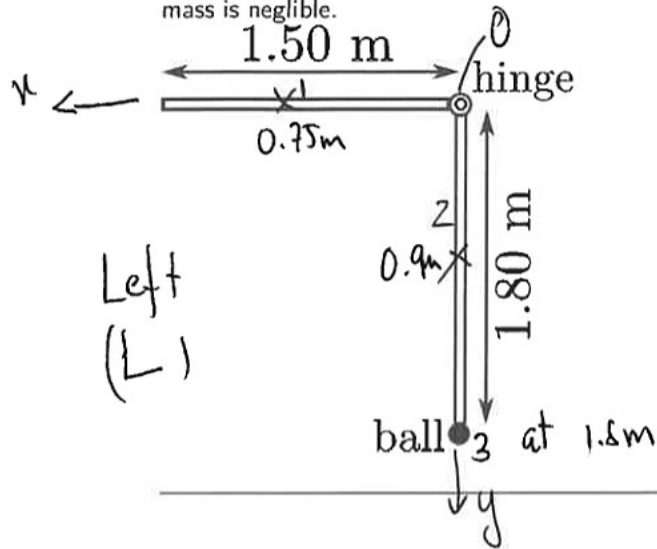
(1)

$$d_2 = \dots = 122.9 \text{ m}$$

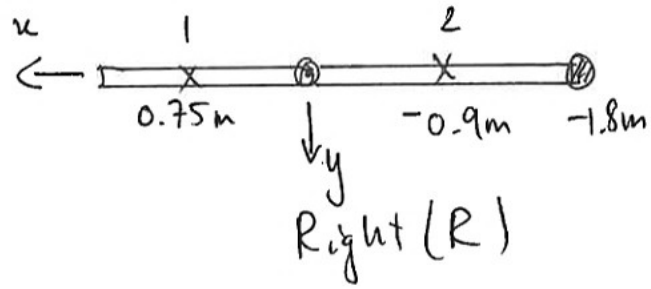
8 Question 8

[11]

A machine part consists of a thin, uniform 4.00 kg bar that is 1.50 m long, hinged perpendicular to a similar vertical bar of mass 3.00 kg and length 1.80 m. The longer bar has a small but dense 2.00 kg ball at one end. By what distance will the center of mass of this machine part move horizontally and vertically if the vertical bar is pivoted counter-clockwise through 90° to make the entire part horizontal? The hinge's mass is negligible.



Left
(L)



Right (R)

$$M = m_1 + m_2 + m_3 = \dots = 9 \text{ kg} \quad \checkmark$$

depends on coordinate system

$$x_{cm,L} = \frac{1}{M} [m_1 x_1 + m_2 x_2 + m_3 x_3] = \dots = 0.333 \text{ m} \quad \checkmark$$

$$y_{cm,L} = \frac{1}{M} [m_1 y_1 + m_2 y_2 + m_3 y_3] = \dots = 0.700 \text{ m} \quad \checkmark$$

Left configuration.

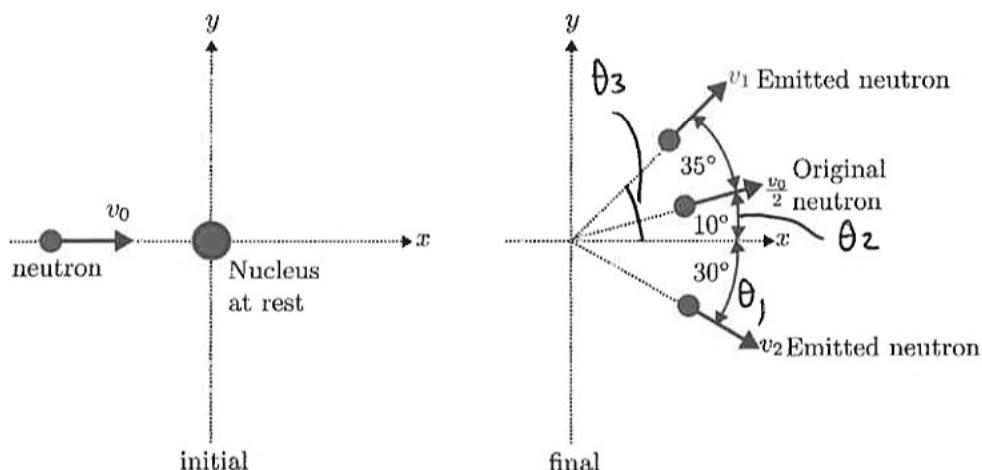
$$x_{cm,R} = \dots = [-0.3667 \text{ m}] = -0.3667 \text{ m} \quad \checkmark$$

$$y_{cm,R} = 0 \quad \checkmark$$

Moves upwards : 0.700 m $\checkmark$ Move to the right : 0.700 m $\checkmark$

Fission, the process that supplies energy in nuclear power plants, occurs when a heavy nucleus is split into two medium-sized nuclei. One such a reaction would occur if a neutron colliding with a ^{235}U nucleus split that nucleus into a ^{141}Ba nucleus and a ^{92}Kr nucleus. In this reaction two neutrons also would be split off from the original ^{235}U .

Before the collision we have the arrangement on the left in the figure below. After the collision we have the Ba nucleus moving in the $+z$ -direction and the Kr nucleus in the $-z$ -direction (not shown in the figure). The three neutrons are moving in the xy -plane as shown on the right of the figure.



- (a) If the incoming neutron has an initial velocity of v_0 , and a final velocity of $v_0/2$ as indicated in the figure, what are the speeds v_1 and v_2 of the other two neutrons in terms of v_0 . (10)

Do momentum conservation:

$$\begin{aligned} x\text{-direction: } p_{ix} &= p_{fx} \\ m_n v_0 &= m_n \frac{v_0}{2} \cos \theta_2 + m_n v_1 \cos \theta_3 + m_n v_2 \cos \theta_1 \\ v_0 &= \dots - (1) \end{aligned}$$

$$\begin{aligned} y\text{-direction: } p_{iy} &= p_{fy} \\ 0 &= m_n \frac{v_0}{2} \sin \theta_2 + m_n v_1 \sin \theta_3 - m_n v_2 \sin \theta_1 \\ v_2 &= \dots - (2) \end{aligned}$$

Eliminate v_2 from (1) using (2):

$$\begin{aligned} v_0 &= \dots \\ &= \dots \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Simplification}$$

$$v_1 = \dots v_0$$

$$v_1 = 0.185 v_0$$

$$v_2 = 0.435 v_0$$

- (b) Find the velocities of the Ba and Kr nuclei in terms of each other. (The mass of the Ba nucleus is approximately 2.3×10^{-25} kg and that of the Kr nucleus is about 1.5×10^{-25} kg.) (3)

Momentum conservation in z-direction: $p_{iz} = p_{fz}$ ✓

$$0 = m_{Ba} v_{Ba} - m_{Kr} v_{Kr}$$

:

$$v_{Kr} = 1.53 v_{Ba} \checkmark$$

$$\text{or } v_{Ba} = \frac{1}{1.53} v_{Kr} = 0.654 v_{Kr}$$

END OF TEST

